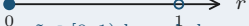
$r \in [0, \infty) \longrightarrow f(r) \in (-\infty, +\infty)$: unbounded input → unbounded output

Encoding

$r \in [0, \infty)$

SP $\tilde{r} = \frac{r}{\sqrt{1+r^2}}$



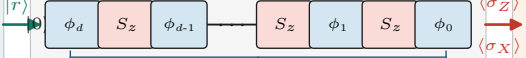
$\tilde{r} \in [0, 1)$ bounded

$$|r\rangle = \frac{r|0\rangle + |1\rangle}{\sqrt{r^2+1}}$$

QSP Circuit

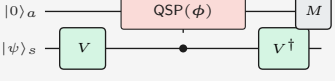
degree d (d+1 phases, d signal operators)

$$S_z = V^{-1} W(\tilde{r})\, V, \quad V = \mathrm{diag}(1, i)$$



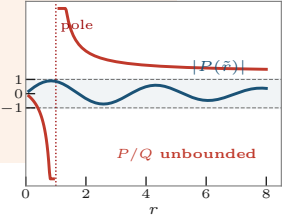
$|P(\tilde{r})|^2 + |Q(\tilde{r})|^2 = 1 \;\Rightarrow\; |P|, |Q| \leq 1$ bounded

Block-encoding context:



Decoding

Pauli measurement

$$f(r) = \frac{P(\tilde{r})}{Q(\tilde{r})}$$

$|P|, |Q| \leq 1$ *bounded*
but P/Q can **diverge**
where $Q(\tilde{r}) = 0$